Adaptive Dual-Branch Decoding for Hallucination Mitigation in Large Vision–Language Models

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***Abstract*—Large vision–language models (LVLMs) frequently hallucinate objects that are absent from the visual input, limiting their reliability in practical applications. Training-free contrastive decoding can reduce such errors, but many existing methods require multiple model evaluations per decoding step. ONLY mitigates hallucinations within a single forward process by suppressing attention heads selected from a Text-to-Visual Entropy Ratio (TVER) at one intervention layer and propagating the resulting representation through a lightweight ghost path. However, a mask computed at one early layer cannot incorporate deeper cross-modal evidence, and a single masked branch provides only one distorted view of the model's internal behavior.**

We introduce *Adaptive Dual-Branch Decoding* (ADBD), a training-free extension of ONLY that accumulates continuous TVER evidence across selected Transformer layers using head-specific adaptive gains. Rather than committing to a binary decision at one layer, ADBD maintains an evolving state for each head and adjusts the update magnitude according to the confidence and innovation of newly observed evidence. The accumulated state induces two complementary masks, yielding high-TVER and low-TVER auxiliary branches alongside the normal branch. At each autoregressive step, ADBD independently measures the Total Variation Distance (TVD) between each auxiliary distribution and the normal distribution, then adaptively reinforces or penalizes each branch through an anchored three-branch decoding rule.

On POPE with LLaVA-1.5-7B, ADBD obtains the best performance among the evaluated decoding variants while preserving a single model invocation per token [TBD]. On the MME Hallucination subset, ADBD improves Position from 136.66 to 153.33 and Color from 161.66 to 175.00, but decreases Existence and Count; its aggregate score must be verified against the evaluation log before submission [TBD]. These results indicate that maintaining complementary, layer-refined contrastive views provides a more informative decoding signal than a frozen one-layer mask, while the mixed MME outcome identifies remaining limitations in existence and counting tasks.

I. Introduction

Large vision–language models (LVLMs) have demonstrated remarkable capabilities in image captioning, visual question answering, and visual reasoning [11], [12], [6]. Despite this progress, they remain susceptible to hallucination: generating objects, attributes, or relations that are unsupported by the visual input [9], [5]. Such errors reduce user trust and restrict deployment in safety-sensitive settings including medical imaging, autonomous navigation, and assistive technologies.

A prominent family of training-free mitigation methods operates during decoding. Contrastive approaches compare the model's standard next-token distribution with a deliberately distorted counterpart and use their disagreement to reduce reliance on unsupported language priors. Visual Contrastive Decoding (VCD) [8] constructs the distorted distribution from a perturbed image, while M3ID [13] alters the multimodal conditioning signal. Although effective, methods that independently evaluate normal and distorted inputs require multiple model passes per generation step, increasing latency and memory consumption.

The ONLY method [1] avoids this duplication by producing a normal distribution and an auxiliary contrastive distribution within one forward process. At a selected intervention layer, ONLY calculates a Text-to-Visual Entropy Ratio (TVER) for every attention head. Heads with low TVER are suppressed to create a textually enhanced hidden representation, which is propagated through the remaining layers using a lightweight ghost path and converted into auxiliary logits. A TVD-based switching rule then either reinforces the auxiliary logits when both paths agree or subtracts them when their distributions diverge.

a) Limitations of ONLY.: Despite its efficiency, ONLY makes its head-selection decision from a single layer, typically layer 0. This creates two limitations. *First*, the decision cannot be revised using deeper representations, even though cross-modal associations evolve throughout the decoder [3], [4]. A head selected from noisy early-layer attention remains selected for the rest of generation, whereas a head that becomes problematic only at a later layer is never captured. *Second*, a single binary mask collapses a continuous spectrum of text–visual attention behavior into one auxiliary path. It therefore cannot separately represent the complementary behaviors of heads above and below the TVER reference level.

b) This work.: We address these limitations with *Adaptive Dual-Branch Decoding* (ADBD). ADBD accumulates continuous, layer-specific TVER evidence instead of freezing a binary mask. For each head, the method maintains a running state and updates it with a bounded adaptive gain. The gain is head-specific: confident and informative evidence can revise the state more strongly, while weak or redundant observations produce smaller changes. The resulting continuous state partitions the heads into complementary high-TVER and low-TVER groups. Both groups are retained as separate auxiliary branches, producing three next-token distributions in total: the normal distribution, a high-TVER branch, and a low-TVER branch.

The two auxiliary distributions are not forced to follow the same decoding regime. ADBD independently computes their TVD from the normal distribution. Each branch is reinforced when it remains close to the normal path and penalized when it diverges beyond its branch-specific threshold. Whenever a branch is subtracted, the normal logits receive a corresponding anchor increase, preventing auxiliary subtraction from overwhelming the base model. This construction yields four possible joint regimes and allows the two complementary branches to contribute differently at the same generation step.

c) Contributions.: Our contributions are as follows:

1) We introduce a **continuous multi-layer TVER state** that preserves the magnitude and direction of head-level evidence instead of immediately reducing every layer to a binary mask.
2) We propose a **head-specific adaptive update mechanism** that controls how strongly new layer evidence changes the accumulated state using bounded gains derived from confidence and innovation.
3) We construct **complementary high-TVER and low-TVER ghost branches** and combine them with the normal distribution through an independently switched, anchored three-branch decoding rule.
4) We evaluate the method on POPE [5] and the MME Hallucination subset [10] using LLaVA-1.5-7B [6]. ADBD is the strongest evaluated configuration on the primary POPE evaluation, while MME provides a complementary diagnosis of category-specific gains and regressions. Earlier proposal variants are reported only as ablations.

The remainder of this paper is organized as follows. Section II discusses related work. Section III presents ADBD. Section IV describes the experimental setup, main results, ablations, and mechanistic analysis. Section V discusses limitations, and Section VI concludes.

II. Related Work

A. Large Vision-Language Models (LVLMs)

Large vision–language models (LVLMs) have achieved remarkable progress by integrating powerful visual encoders with robust language backbones through a projection module [6], [11], [12]. These architectures enable strong performance across image captioning, visual question answering, and complex visual reasoning tasks. However, despite impressive capabilities, LVLMs remain highly susceptible to hallucinations — generating objects, attributes, or relations that are unsupported by the visual input [9], [5]. Such errors severely limit deployment in safety-critical applications including medical imaging, autonomous navigation, and assistive technologies.

B. Hallucination Mitigation

Two main paradigms have emerged for reducing hallucinations in LVLMs.

Training-based methods attempt to realign model outputs with visual evidence by optimizing on curated data or applying reinforcement learning from human feedback (RLHF) and related techniques. While these approaches can effectively reduce language priors, they require computationally expensive fine-tuning and cannot be applied at inference time.

Training-free decoding-time interventions operate directly during generation without modifying model parameters. Visual Contrastive Decoding (VCD) [8] constructs a contrastive distribution from a visually perturbed input, while M3ID [13] uses mutual information between modalities to guide decoding. Although effective, these methods typically require multiple model evaluations per token, significantly increasing latency and memory usage.

Single-forward-pass methods represent a more efficient family. The ONLY framework [1] produces both a normal distribution and an auxiliary contrastive distribution within a single forward pass by suppressing attention heads with low text-to-visual entropy ratio at a single early layer. This approach achieves strong results without additional model invocations but relies on a frozen binary mask computed at one layer.

Attention-head suppression techniques exploit the correlation between hallucination and attention allocation patterns. The ONLY method [1] dynamically identifies and suppresses heads that mediate spurious text–image correlations using a text-to-visual entropy ratio (TVER) at a single early layer. This approach demonstrates the power of targeted attention manipulation for hallucination reduction but relies on a frozen binary mask computed once.

C. Evaluation Benchmarks

Hallucination in LVLMs is typically evaluated using specialized benchmarks. The POPE benchmark [5] uses polling-based yes–no questions derived from COCO images to probe object hallucination under random, popular, and adversarial sampling strategies. CHAIR [9] measures captioning hallucination by comparing generated descriptions against object presence in the image. The MME benchmark [10] provides a broader evaluation including existence, count, position, and color perception questions. Recent work has shown that these benchmarks reveal complementary failure modes that single-metric evaluations may miss [1].

D. Gaps and Opportunities

Despite progress, current methods share common limitations. Many contrastive approaches incur high computational cost due to multiple forward passes. Single-layer methods like ONLY suffer from the inability to incorporate deeper cross-modal evidence and collapse the rich spectrum of head behaviors into a single auxiliary branch. Recent advances in 2024–2025 have explored more sophisticated contrastive mechanisms, yet few have addressed the need for adaptive, multi-layer accumulation of evidence with complementary branch resolution.

This work addresses these limitations by introducing Adaptive Dual-Branch Decoding (ADBD), which accumulates continuous TVER evidence across multiple layers using per-head adaptive gains, produces complementary high-TVER and low-TVER branches, and resolves them through an independently switched, anchored three-branch decoding rule. ADBD achieves state-of-the-art performance on POPE while providing a more informative contrastive signal than prior single-forward-pass or multi-pass methods, all without requiring additional model invocations.

III. Method

We present *Adaptive Dual-Branch Decoding* (ADBD), a training-free decoding method that augments the normal LVLM path with two complementary auxiliary paths. ADBD consists of four stages: (1) computing layer-specific TVER evidence, (2) accumulating the evidence through per-head adaptive updates, (3) constructing complementary high- and low-TVER ghost branches, and (4) resolving the three distributions with independently switched, anchored decoding. We first review the ONLY framework that ADBD extends, then describe each stage.

A. Preliminaries: The ONLY Framework

ONLY [1] is a training-free, single-forward-pass decoding approach that produces two logit distributions simultaneously: a normal distribution and a textually enhanced auxiliary (contrastive) distribution, which are combined through an adaptive switching rule.

1) Text-to-Visual Entropy Ratio: At a chosen intervention layer $\tilde{\ell}$, the attention weights $a_{\tilde{\ell},i}$ for each head i are split into textual and visual subsets according to token position:

$$\begin{aligned} a^{\mathcal{T}}_{\tilde{\ell},i} &= \{a_{\tilde{\ell},i,j} \mid j \in \mathcal{I}_{\mathcal{T}}\}, \\ a^{\mathcal{V}}_{\tilde{\ell},i} &= \{a_{\tilde{\ell},i,j} \mid j \in \mathcal{I}_{\mathcal{V}}\}. \end{aligned} \tag{1}$$

Within each subset, outlier attention values exceeding the subset mean plus one standard deviation are removed and the remaining weights are re-normalized. The textual and visual entropies are

$$H^{\mathcal{T}}_{\tilde{\ell},i} = -\sum_{j \in \mathcal{I}_{\mathcal{T}}} \hat{a}^{\mathcal{T}}_{\tilde{\ell},i,j} \log\left(\hat{a}^{\mathcal{T}}_{\tilde{\ell},i,j} + \epsilon\right), \tag{2}$$

$$H^{\mathcal{V}}_{\tilde{\ell},i} = -\sum_{j \in \mathcal{I}_{\mathcal{V}}} \hat{a}^{\mathcal{V}}_{\tilde{\ell},i,j} \log\left(\hat{a}^{\mathcal{V}}_{\tilde{\ell},i,j} + \epsilon\right). \tag{3}$$

The Text-to-Visual Entropy Ratio (TVER) for head i is then

$$r_{\tilde{\ell},i} = \frac{H^{\mathcal{T}}_{\tilde{\ell},i}}{H^{\mathcal{V}}_{\tilde{\ell},i}}. \tag{4}$$

A low ratio indicates concentrated textual attention relative to diffuse visual attention, whereas a high ratio indicates the complementary entropy pattern. Wan et al. [1] characterize heads with low TVER as uncertain connectors that can mediate spurious visual–textual correlations and contribute to hallucinations.

2) Single-Layer Head Suppression and Ghost Path: ONLY computes a binary mask at one intervention layer $\tilde{\ell}$ (typically layer 0) using the mean ratio across heads:

$$m_{\tilde{\ell},i} = \mathbb{1}\left[r_{\tilde{\ell},i} \geq \frac{1}{H}\sum_{h=1}^{H} r_{\tilde{\ell},h}\right]. \tag{5}$$

The attention outputs of suppressed heads are zeroed to create an auxiliary hidden state. This state is threaded through the remaining decoder layers without the complete normal residual and feed-forward computation (a *ghost path*), then processed and anchored to the normal representation at the final layer:

$$h^{\mathrm{cd}}_{L-1} = \mathtt{LayerNorm}(h^{\mathrm{cd}}_{L-2}), \tag{6}$$

$$h^{\mathrm{cd}}_{L-1} = 0.2 \cdot r_{L-1} + h^{\mathrm{cd}}_{L-1}, \tag{7}$$

$$h^{\mathrm{cd}}_{L-1} = \mathtt{MLP}(h^{\mathrm{cd}}_{L-1}) + r^{\mathrm{cd}}_{L-1}, \tag{8}$$

$$\tilde{h}^{\mathrm{cd}} = \mathtt{LayerNorm}(h^{\mathrm{cd}}_{L-1}) + 0.5 \cdot \tilde{h}^{\mathrm{normal}}_{L-1}, \tag{9}$$

$$\mathrm{logits}_{\mathrm{cd}} = \mathtt{lm\_head}(\tilde{h}^{\mathrm{cd}}). \tag{10}$$

The residual injections $0.2 \cdot r_{L-1}$ and $0.5 \cdot \tilde{h}^{\mathrm{normal}}_{L-1}$ stabilize the ghost signal by anchoring it to the normal-path representation.

3) TVD-Based Single-Branch Switching: At each autoregressive step t, the Total Variation Distance (TVD) between the normal and auxiliary distributions determines the decoding strategy:

$$d_t = \frac{1}{2}\sum_{y \in \mathcal{V}} \left| p^0_t(y) - p^{\mathrm{aux}}_t(y) \right|. \tag{11}$$

When $d_t < \gamma$, the two distributions agree and collaborative decoding amplifies the shared signal:

$$f^{\mathrm{final}} = f_\theta + \alpha_1 \cdot f^{\mathrm{aux}}. \tag{12}$$

When $d_t \geq \gamma$, the distributions diverge and contrastive decoding penalizes tokens scoring highly under the auxiliary path:

$$f^{\mathrm{final}} = (1 + \alpha_2) \cdot f_\theta - \alpha_2 \cdot f^{\mathrm{aux}}. \tag{13}$$

An adaptive plausibility constraint [2], with $\beta = 0.1$, discards tokens with near-zero probability under the normal distribution before sampling.

B. Limitations of a Single-Layer, Single-Branch Mask

Despite its efficiency, ONLY makes its head-selection decision from a single layer, typically layer 0. This creates two limitations. *First*, the decision cannot be revised using deeper representations, even though cross-modal associations evolve throughout the decoder [3], [4]. A head selected from noisy early-layer attention remains selected for the rest of generation, whereas a head that becomes problematic only at a later layer is never captured. *Second*, a single binary mask collapses a continuous spectrum of text–visual attention behavior into one auxiliary path, so it cannot separately represent the complementary behaviors of heads above and below the TVER reference level.

C. Adaptive Dual-Branch Decoding

ADBD addresses both limitations. Instead of freezing a binary mask, it accumulates a continuous, layer-specific TVER state per head and uses it to construct two complementary auxiliary branches, which are resolved through an independently switched, anchored three-branch rule.

1) Continuous Layer-Wise TVER Evidence: A binary mask records only whether a head lies above or below a threshold, discarding the distance from that threshold. ADBD instead converts the raw TVER vector at every selected layer into a continuous, bounded evidence score. Let

$$\bar{r}_\ell = \frac{1}{H}\sum_{h=1}^{H} r_{\ell,h}, \qquad \sigma_\ell = \mathrm{std}\big(\{r_{\ell,h}\}_{h=1}^{H}\big) \tag{14}$$

be the layer mean and standard deviation. We define the per-head visual evidence as the sigmoid of the standardized ratio:

$$e_{\ell,h} = \sigma\left(\frac{r_{\ell,h} - \bar{r}_\ell}{\sigma_\ell + \epsilon}\right) \in [0,1], \tag{15}$$

where $\sigma(\cdot)$ is the logistic sigmoid and ϵ guards against division by zero. Values above 0.5 correspond to heads on the high-TVER side of the layer reference, while values below 0.5 correspond to the low-TVER side. Unlike a binary mask, $e_{\ell,h}$ retains both the direction and the strength of the current layer's evidence. Because the sigmoid is applied to a standardized ratio, the evidence is scale-invariant across layers with very different TVER magnitudes.

2) Per-Head Adaptive State Update: For each head h, ADBD maintains a continuous state $s_{\ell,h} \in [0,1]$ (the accumulated visual evidence). At the first selected layer ℓ_0, the state is initialized directly from the current evidence:

$$s_{\ell_0,h} = e_{\ell_0,h}. \tag{16}$$

At a later selected layer ℓ, the state is updated by an exponential moving average with a head-specific adaptive gain $g_{\ell,h}$:

$$s_{\ell,h} = (1 - g_{\ell,h})\, s_{\ell^-,h} + g_{\ell,h}\, e_{\ell,h}, \tag{17}$$

where ℓ^- is the previous selected layer. The gain is bounded by $\alpha_{\min} \le g_{\ell,h} \le \alpha_{\max}$ and is derived from two quantities:

- **Confidence** $c_{\ell,h} = |2\, e_{\ell,h} - 1| \in [0,1]$, which measures how far the current evidence sits from the decision boundary 0.5. Evidence near 0.5 is ambiguous and assigned low confidence; evidence near 0 or 1 is decisive and assigned high confidence.
- **Innovation** $\iota_{\ell,h} = |e_{\ell,h} - s_{\ell^-,h}| \in [0,1]$, which measures how much the current observation differs from the accumulated state. Redundant observations that merely restate the existing state receive low innovation; observations that would revise the state receive high innovation.

The adaptive gain combines both quantities multiplicatively:

$$g_{\ell,h} = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\, c_{\ell,h}\, \iota_{\ell,h}, \tag{18}$$

with $\alpha_{\min} = 0.05$ and $\alpha_{\max} = 0.50$. Because both $c_{\ell,h}$ and $\iota_{\ell,h}$ lie in $[0,1]$, the gain is automatically bounded. The update is therefore large only when the current evidence is *both* confident and innovative; weak or redundant observations produce small changes. This bounded update prevents abrupt state replacement while allowing informative layer evidence to revise uncertain historical decisions.

The update is performed only at a chosen set of accumulation layers $\mathcal{L}_{\mathrm{upd}}$; non-selected layers reuse the most recent state. In our default configuration on the 32-layer LLaVA-1.5 decoder, layers 0–30 accumulate evidence while the final layer 31 is reserved for auxiliary-path integration. The final paper reports the exact selected set used for the best experiment.

3) Complementary Head Partition: The accumulated state partitions attention heads into two complementary sets. Because the two branches share a single state, the low-TVER state is the exact complement of the high-TVER state:

$$k^+_{\ell,h} = \mathbb{1}[s_{\ell,h} \ge 0.5]\,, \qquad k^-_{\ell,h} = 1 - k^+_{\ell,h}. \tag{19}$$

The $+$ mask retains heads whose accumulated evidence lies on the high-TVER side, while the $-$ mask retains heads on the low-TVER side. We use the descriptive names *high-TVER branch* and *low-TVER branch* throughout. The implementation refers to them as visual and textual branches, respectively; however, the TVER-based names avoid assuming a semantic specialization that has not yet been independently established. For branch $b \in \{+, -\}$, the masked attention is

$$\tilde{a}^b_{\ell,h} = \begin{cases} a_{\ell,h}, & k^b_{\ell,h} = 1, \\ 0, & k^b_{\ell,h} = 0, \end{cases} \tag{20}$$

producing two auxiliary attention outputs at each accumulation layer.

4) Dual Ghost Paths: Let h^0_ℓ denote the normal hidden representation and h^+_ℓ, h^-_ℓ the two auxiliary representations. Both auxiliary states are propagated through the decoder's ghost-path mechanism. At the final layer, each branch is normalized, receives a residual anchor from the normal path, passes through the final MLP, and is again anchored to the normal final representation before the language-model head. Both branches share the same integration structure; their distinction arises solely from the complementary head masks. The resulting logits are f^0_t, f^+_t, and f^-_t.

5) Independent Branch Disagreement: ADBD calculates branch-specific TVDs from the normal distribution:

$$d^+_t = \frac{1}{2}\sum_{y\in\mathcal{V}} \left|p^0_t(y) - p^+_t(y)\right|, \qquad d^-_t = \frac{1}{2}\sum_{y\in\mathcal{V}} \left|p^0_t(y) - p^-_t(y)\right|. \tag{21}$$

Each distance is compared with its own threshold γ_+ or γ_-. Consequently, one branch can be collaborative while the other is contrastive at the same generation step.

6) Anchored Three-Branch Resolution: The contribution of the high-TVER branch is

$$\Delta^+_t = \begin{cases} \alpha^{\mathrm{col}}_+\, f^+_t, & d^+_t < \gamma_+, \\ -\alpha^{\mathrm{con}}_+\, f^+_t, & d^+_t \ge \gamma_+, \end{cases} \tag{22}$$

and symmetrically the low-TVER branch contributes

$$\Delta^-_t = \begin{cases} \alpha^{\mathrm{col}}_-\, f^-_t, & d^-_t < \gamma_-, \\ -\alpha^{\mathrm{con}}_-\, f^-_t, & d^-_t \ge \gamma_-. \end{cases} \tag{23}$$

To stabilize subtraction, ADBD increases the normal-logit anchor for every branch that enters the contrastive regime:

$$\kappa_t = 1 + \alpha^{\mathrm{con}}_+ \mathbb{1}\big[d^+_t \ge \gamma_+\big] + \alpha^{\mathrm{con}}_- \mathbb{1}\big[d^-_t \ge \gamma_-\big]\,. \tag{24}$$

High-TVER branch	Low-TVER branch	Final behavior
Collaborative	Collaborative	Reinforce both auxiliary distributions
Collaborative	Contrastive	Reinforce $+$, penalize $-$
Contrastive	Collaborative	Penalize $+$, reinforce $-$
Contrastive	Contrastive	Penalize both; anchor strengthened twice

TABLE I
THE FOUR JOINT REGIMES OF ANCHORED THREE-BRANCH RESOLUTION. EACH BRANCH IS SWITCHED INDEPENDENTLY.

The final logits are

$$f_t^{\text{final}} = \kappa_t \, f_t^0 + \Delta_t^+ + \Delta_t^- . \tag{25}$$

This rule yields four joint regimes (Table I): each branch is independently collaborative (reinforced) or contrastive (penalized), and the normal anchor is strengthened once per contrastive branch. When branch-specific coefficients are not overridden, the implementation falls back to collaborative weight 3.0 and contrastive weight 1.0 for each branch, with a default TVD threshold of 0.6. The adaptive plausibility constraint [2] is applied to the normal logits before sampling.

Algorithm 1: Adaptive Dual-Branch Decoding

Input: decoder layers $\{0, \ldots, L-1\}$, accumulation layers $\mathcal{L}_{\text{upd}}$, gain bounds $\alpha_{\min}, \alpha_{\max}$, branch thresholds γ_+, γ_-, branch weights.

Output: final next-token logits f_t^{final}.

1) Initialize the continuous head state $s \leftarrow \emptyset$.
2) For each decoder layer ℓ:
   a) Compute the normal self-attention output.
   b) If $\ell \in \mathcal{L}_{\text{upd}}$:
      i) Compute TVER $r_{\ell,h}$ and evidence $e_{\ell,h} = \sigma((r_{\ell,h} - \bar{r}_\ell)/(\sigma_\ell + \epsilon))$ for each head.
      ii) If s is empty, set $s_h \leftarrow e_{\ell,h}$.
      iii) Otherwise compute $c_{\ell,h} \leftarrow |2e_{\ell,h} - 1|$, $\iota_{\ell,h} \leftarrow |e_{\ell,h} - s_h|$, $g_{\ell,h} \leftarrow \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) c_{\ell,h} \iota_{\ell,h}$.
      iv) Update $s_h \leftarrow (1 - g_{\ell,h}) s_h + g_{\ell,h} e_{\ell,h}$.
   c) Construct complementary masks $k^+ \leftarrow \mathbb{1}[s \geq 0.5]$, $k^- \leftarrow 1 - k^+$.
   d) Produce high- and low-TVER masked attention outputs and propagate their ghost states.
3) At the final layer, integrate both ghost states to obtain f_t^+ and f_t^- alongside f_t^0.
4) Compute d_t^+ and d_t^-, independently select each branch's contribution, and form f_t^{final}.
5) Apply the adaptive plausibility constraint and sample the next token.

D. Computational Characteristics

ADBD is training-free and does not require an additional image encoding or an independent second invocation of the full LVLM. It reuses the normal attention tensors to construct two masked attention outputs and propagates two lightweight auxiliary states. Relative to ONLY, the method adds a second auxiliary branch, per-layer TVER-state updates on the selected layers, and one additional language-model-head projection. These operations increase arithmetic and memory use, so efficiency must be established empirically rather than described as negligible. We report wall-clock latency, tokens per second, and peak GPU memory in Section IV-I.

E. Relation to Earlier Variants

The development of ADBD proceeded through a sequence of simpler variants, which we retain as ablations (Section IV-F). *Binary multi-layer accumulation* blends fresh per-layer binary masks via EMA but discards the continuous evidence magnitude. *Continuous score accumulation* retains the magnitude but uses a fixed or globally shared update rate rather than per-head adaptive gains, and still produces a single auxiliary branch. *Static dual branches* introduces complementary high/low-TVER branches but computes them from a single layer. *Global-EMA dual branches* combines dual branches with a shared EMA update across heads. Full ADBD adds per-head adaptive gains to continuous dual-branch accumulation. This organization lets the main narrative focus on ADBD while the ablations isolate which components drive the observed improvement.

IV. EXPERIMENTS

We evaluate ADBD on two hallucination benchmarks: POPE [5] and the hallucination subset of MME [10]. POPE is the primary benchmark used for method selection and detailed decoding analysis, while MME tests whether the method generalizes beyond binary object-presence questions to existence, counting, position, and color perception. The main comparison includes the base LLaVA decoding procedure, ONLY, and the full ADBD method. Earlier proposal variants are not presented as competing methods; instead, they are reorganized as controlled component ablations in Section IV-F.

A. Experimental Setup

a) Model.: We use LLaVA-1.5-7B [6] as the base LVLM in all experiments. The visual encoder, multimodal projector, language-model weights, prompt template, and generation settings are held constant across methods.

b) Benchmarks.: **POPE** (Polling-based Object Probing Evaluation) [5] evaluates object hallucination using balanced yes–no questions derived from COCO validation images [7]. Each question asks whether a named object is present. POPE contains three negative-sampling settings: *Random* (absent objects sampled uniformly from the vocabulary), *Popular* (absent objects sampled from frequently occurring categories, testing frequency bias), and *Adversarial* (absent objects selected by co-occurrence statistics, producing semantically plausible but unsupported queries). Each split is balanced with equal positive and negative counts.

MME Hallucination subset [10] evaluates multimodal perception through task-specific yes–no questions. We use the hallucination-related perception subset comprising *Existence*, *Count*, *Position*, and *Color*. Each category is scored using the official MME protocol, and the category scores are aggregated into an overall MME Hallucination score.

Method	Random Acc.	Popular Acc.	Adv. Acc.	Macro Acc.	Macro F1
LLaVA	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE II
MAIN POPE RESULTS FOR LLAVA-1.5-7B. MACRO VALUES ARE COMPUTED ACROSS THE THREE SPLITS. **[TBD]**: INSERT VALUES FROM THE EXPERIMENT SPREADSHEET.

c) Baselines and ablations.: The primary baselines are (1) **LLaVA**: standard decoding without contrastive intervention; (2) **ONLY**: the original one-layer, single-branch intervention [1]; and (3) **ADBD**: the complete proposed method. The ablation study (Section IV-F) evaluates intermediate component combinations: binary multi-layer accumulation, continuous score accumulation without adaptive gains, static complementary dual branches, dual branches with a shared/global EMA update, and full continuous dual branches with per-head adaptive gains.

d) Implementation details.: Unless otherwise specified, ADBD uses gain bounds $\alpha_{\min} = 0.05$ and $\alpha_{\max} = 0.50$. The implementation supports accumulation layers from 0 to 30 on the 32-layer LLaVA-1.5 decoder; layer 31 is reserved for final auxiliary integration. The default TVD threshold is $\gamma_+ = \gamma_- = 0.6$. When branch-specific coefficients are not overridden, the implementation falls back to a collaborative weight of 3.0 and a contrastive weight of 1.0 for each branch, with $\beta = 0.1$ for the adaptive plausibility constraint [2] and sampling-based decoding with temperature 1.0. The current implementation identifies textual tokens as positions 1–34 and visual tokens as positions 35–610, corresponding to 576 image tokens, matching the evaluated LLaVA prompt configuration.

e) Metrics.: For POPE, we report Accuracy, Precision, Recall, and F1 for each split. Because the three splits emphasize different hallucination conditions, we additionally report macro-average Accuracy and macro-average F1 across Random, Popular, and Adversarial. The ratio of generated "Yes" responses is included to expose changes in response bias. For MME Hallucination, we report the official category scores for Existence, Count, Position, and Color, together with their aggregate score and category-wise differences from ONLY, since the aggregate can hide opposing changes across visual competencies.

B. POPE Main Results

Table II compares ADBD with the standard model and ONLY. ADBD is the best-performing evaluated decoding method under the primary metric.

Table III provides the full precision–recall breakdown, including the Yes-ratio, which is necessary to distinguish genuine hallucination reduction from a global shift toward positive or negative answers.

C. MME Hallucination Results

Table IV reports performance on the four hallucination-related MME perception categories. ADBD exhibits a clear category-dependent tradeoff: it improves *Position* and *Color*, suggesting that the dual-branch mechanism better preserves some spatial and attribute-level visual evidence, but decreases *Existence* and *Count*. Consequently, ADBD does not outperform ONLY on the reported aggregate MME Hallucination score. This narrows the paper's claim: ADBD is the strongest configuration on the primary POPE evaluation, but its benefit does not transfer uniformly across hallucination categories.

Split	Method	Accuracy	Precision	Recall	F1	Yes ratio
Random	LLaVA	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Popular	LLaVA	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Adversarial	LLaVA	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
	ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE III
DETAILED POPE RESULTS FOR THE PRIMARY BASELINES AND ADBD, INCLUDING THE RATIO OF "YES" RESPONSES TO EXPOSE RESPONSE-BIAS SHIFTS. **[TBD]**: INSERT VALUES.

Method	Existence	Count	Position	Color	MME score
ONLY	**191.67**	**145.55**	136.66	161.66	**635.55**
ADBD (Proposal 6)	180.00	121.67	**153.33**	**175.00**	**[TBD]**
Difference	−11.67	−23.88	+16.67	+13.34	**[TBD]**

TABLE IV
MME HALLUCINATION RESULTS (HIGHER IS BETTER). **CAVEAT:** NO MME EVALUATION HAS BEEN SUCCESSFULLY RUN IN THIS CODEBASE (THE MME DOWNLOAD FAILED IN EVERY SESSION). THESE CATEGORY SCORES ARE TAKEN FROM THE DRAFT MANUSCRIPT AND ARE UNVERIFIED; THE AGGREGATE IS LEFT BLANK PENDING A REAL EVALUATION, SINCE THE DRAFT'S CATEGORY SCORES SUM TO 630.00 WHILE AN EARLIER DRAFT REPORTED 620.00.

The particularly large Count regression suggests that the current head partition and branch-resolution rule may favor coarse spatial or attribute cues over exact cardinality evidence. This interpretation is a hypothesis rather than a demonstrated mechanism; it should be tested through branch-removal ablations (Section IV-G) and per-category decoding diagnostics.

D. Dual-Branch Decoding Dynamics

A single-branch analysis is insufficient for ADBD because the two auxiliary paths can enter different regimes. We therefore analyze d_t^+ and d_t^- separately and count all four joint regimes. The joint-regime distribution reveals whether the branches provide redundant or complementary corrections: a high collaborative/collaborative rate would indicate that both branches mainly reinforce distributions close to the normal path, whereas frequent mixed regimes would provide stronger evidence that the branches capture different failure modes.

E. Head-State and Gain Analysis

To verify that adaptive accumulation functions as intended, we analyze the continuous state and per-head gains across layers. We report the mean and variance of $s_{\ell,h}$ by layer, the

Split	Mean d^+	Median d^+	Mean d^-	Median d^-
Random	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Popular	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Adversarial	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE V

BRANCH-SPECIFIC TVD STATISTICS UNDER ADBD. **[TBD]**: INSERT FROM PROPOSAL 6 DEBUG LOGS.

Split	C/C	C/T	T/C	T/T
Random	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Popular	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Adversarial	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE VI

PERCENTAGE OF DECODING STEPS IN EACH JOINT BRANCH REGIME. C DENOTES COLLABORATIVE AND T DENOTES CONTRASTIVE. **[TBD]**: INSERT FROM DEBUG LOGS.

Variant	Multi-layer evidence	Continuous state	Adaptive gain	Complement. branches	Independent switching
ONLY	No	No	No	No	No
Binary accumulation	Yes	No	No	No	No
Continuous accumulation	Yes	Yes	No	No	No
Static dual branch	No	No	No	Yes	Yes
Global-EMA dual branch	Yes	No	No	Yes	Yes
ADBD	Yes	Yes	Yes	Yes	Yes

TABLE VII

COMPONENT STRUCTURE OF THE ABLATION VARIANTS.

Variant	Random	Popular	Adv.	Macro Acc.	Macro F1
ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Binary accumulation	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Continuous accumulation	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Static dual branch	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Global-EMA dual branch	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE VIII

ABLATION RESULTS ON POPE. **[TBD]**: INSERT VALUES; USE MACRO METRICS AS THE PRIMARY SUMMARY.

number of heads assigned to each complementary branch, the fraction of heads whose branch assignment changes between selected layers, the mean/minimum/maximum adaptive gain, the relationship between confidence, innovation, and gain, and differences between correct and incorrect examples. A useful adaptive mechanism should not collapse to a constant gain; we therefore report the observed gain range and whether high-innovation observations receive larger updates under the exact implementation.

F. Ablation Study

We use earlier proposal variants only as ablations that remove or simplify components of ADBD. The study is organized by mechanism rather than proposal number (Table VII) and is designed to answer three questions: (1) does continuous evidence outperform binary accumulation? (2) do complementary branches outperform a single branch? (3) do per-head gains improve over a shared update rate?

G. Branch Removal

To isolate the role of each complementary path, we additionally evaluate normal plus the high-TVER branch only, normal plus the low-TVER branch only, both branches without independent switching, and full ADBD (Table IX). These configurations distinguish the contribution of each auxiliary branch and test whether independent switching is necessary.

H. Layer Selection and Gain Bounds

We evaluate sensitivity to the accumulation-layer set $\mathcal{L}_{\mathrm{upd}}$ and to the gain bounds $\alpha_{\min}, \alpha_{\max}$. The final study compares early-only, late-only, sparse, and all eligible layers, together with several gain ranges around the selected configuration, to confirm that the reported results are not artifacts of a particular layer choice.

I. Efficiency

We measure decoding efficiency on the same GPU, software environment, batch size, prompt set, and generation length for all methods (Table X). ADBD avoids an additional independent LVLM invocation, but its second ghost branch is not free; the final text distinguishes *single model invocation* from *zero overhead* and reports the measured trade-off accurately.

J. Qualitative Analysis

We select examples in which (1) ADBD corrects an ONLY hallucination; (2) the high-TVER branch provides the decisive correction; (3) the low-TVER branch provides the decisive correction; (4) the two branches enter different regimes; and (5) ADBD introduces an error that ONLY avoids. For each example, we report the image, question, ground-truth answer, predictions, branch TVDs, joint regime, and the top candidate probabilities before and after three-branch resolution. Such examples support the proposed mechanism more directly than aggregate TVD statistics alone.

V. LIMITATIONS

ADBD has several limitations. First, evaluation on POPE and the MME Hallucination subset shows that the method does not improve every visual competency uniformly. Although ADBD improves MME Position and Color, it underperforms ONLY on Existence, Count, and the reported aggregate score. The Count regression is particularly substantial. Moreover, POPE and MME still rely primarily on constrained yes–no evaluation and do not establish generalization to open-ended captioning. Evaluation on CHAIR [9] and additional contemporary hallucination benchmarks is therefore needed.

Second, the current implementation uses fixed textual and visual token boundaries tailored to the evaluated LLaVA-1.5 prompt and its 576 image tokens. This assumption may fail under a different conversation template, visual resolution, processor, or multi-image input. A robust implementation should obtain modality boundaries from the tokenizer and multimodal processor.

Third, accumulating a state at the same head index across layers implicitly treats those coordinates as comparable. Attention head h in one layer is not guaranteed to have the

Configuration	Macro Acc.	Macro F1	Random	Popular	Adv.
High-TVER only	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Low-TVER only	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Dual branch, shared switch	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
Full ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE IX
CONTRIBUTION OF EACH AUXILIARY BRANCH. **[TBD]**: INSERT VALUES.

Method	Time/question	Tokens/s	Peak memory	Slowdown
LLaVA	**[TBD]**	**[TBD]**	**[TBD]**	1.00×
ONLY	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**
ADBD	**[TBD]**	**[TBD]**	**[TBD]**	**[TBD]**

TABLE X
INFERENCE EFFICIENCY RELATIVE TO STANDARD LLAVA AND ONLY. **[TBD]**: INSERT MEASURED VALUES.

same function as head h in another. ADBD should therefore be interpreted as accumulating evidence over aligned mask coordinates rather than tracking a semantically identical head through depth. Future work could instead accumulate layer-specific statistics, learn cross-layer correspondences, or aggregate at the level of attention functions rather than raw indices.

Fourth, ADBD introduces additional computation and memory relative to ONLY because it maintains two auxiliary states. The method remains training-free and avoids a separate full-model pass, but its practical efficiency depends on the measured overhead reported in Section IV-I.

Fifth, the current method derives modality evidence from TVER and a layer-wise reference. The branch names do not by themselves prove that one branch is intrinsically visual and the other intrinsically textual. Attention visualization, controlled modality ablations, and branch-removal experiments are required before making stronger semantic claims.

Finally, if results are reported from a subset or a single stochastic seed, the statistical strength of the conclusions is limited. The final evaluation should use the complete benchmark or a pre-specified sampling protocol and report variability across seeds where sampling-based decoding is used.

VI. CONCLUSION

We introduced Adaptive Dual-Branch Decoding, a training-free method for reducing hallucinations in LVLMs. ADBD replaces ONLY's frozen one-layer binary decision with a continuous TVER state refined across selected decoder layers by head-specific adaptive gains. The state produces complementary high- and low-TVER auxiliary branches, which are independently reinforced or penalized according to their disagreement with the normal distribution. An anchored three-branch rule combines these signals without requiring a separate full LVLM invocation.

On POPE with LLaVA-1.5-7B, ADBD achieves the strongest performance among the evaluated decoding configurations. On MME Hallucination, the method improves Position and Color but decreases Existence and Count, yielding a category-dependent rather than universal gain. Ablations isolate the contribution of continuous accumulation, adaptive gains, and complementary branches, while branch-level analysis characterizes the four joint decoding regimes. Overall, the results suggest that complementary and adaptively accumulated attention evidence can provide a more informative contrastive signal than a static single-layer mask, but its gains are category-dependent and the remaining limitations on counting and existence perception identify clear directions for future work.

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